

Problem set 1: Lab: Xv6 and Unix utilities

Sleep

```
maskatia@LAPTOP-S8TAA5HI X + v
builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-b
uiltin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o user/find.o user/find.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_find user/find.o user/ulib.o user/usys.o user/printf.o user/umall
oc.o
riscv64-linux-gnu-objdump -S user/_find > user/find.asm
riscv64-linux-gnu-objdump -t user/_find | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/find.sym
make: *** No rule to make target 'user/uptime.c', needed by 'user/uptime.o'. Stop.
FatimaFaisal@xv6:~$ nano Makefile
FatimaFaisal@xv6:~$ make qemu
mv -f fs.img fs.img.bk
mv: cannot stat 'fs.img': No such file or directory
make: [Makefile:290: newfs.img] Error 1 (ignored)
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_
ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forp
han user/_dorphan user/_sleep user/_sixfive user/_memdump user/_find user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1102 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -
drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sleep
Usage: sleep <ticks>
$ sleep 100
$ |
```

Sleep grade test:

```
maskatia@LAPTOP-S8TAA5HI X + v
ng -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-b
uiltin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-b
uiltin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o kernel/console.o kernel/console.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mmodel=medany -ffreestandi
ng -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-b
uiltin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-b
uiltin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o kernel/printf.o kernel/printf.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mmodel=medany -ffreestandi
ng -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-b
uiltin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-b
uiltin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o kernel/uart.o kernel/uart.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mmodel=medany -ffreestandi
ng -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-b
uiltin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-b
uiltin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel
/main.o kernel/vm.o kernel/proc.o kernel/switch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o ker
nel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plio.o
kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/maskatia/xv6-labs-2025'
== Test sleep, no arguments == sleep, no arguments: OK (4.0s)
== Test sleep, returns == sleep, returns: OK (0.8s)
== Test sleep, makes syscall == sleep, makes syscall: OK (1.0s)
FatimaFaisal@xv6:~$
```

Sixfive

```
FatimaFaisal@xv6:~$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_
ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forp
han user/_dorphan user/_sleep user/_sixfive user/_memdump user/_find user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1102 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -
drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sixfive
Usage: sixfive <file>...
$ sixfive sixfive.txt
5
100
18
6
$ |
```

Sixfive grade test:

```
builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-b
uiltin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o kernel/console.o kernel/console.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestandi
ng -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-
builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-b
uiltin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o kernel/printf.o kernel/printf.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestandi
ng -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-
builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-b
uiltin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o kernel/uart.o kernel/uart.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestandi
ng -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-
builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-b
uiltin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel
/main.o kernel/vm.o kernel/proc.o kernel/switch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o ker
nel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o
kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/maskatia/xv6-labs-2025'
== Test sixfive_test == sixfive_test: OK (3.8s)
== Test sixfive_readme == sixfive_readme: OK (1.2s)
== Test sixfive_all == sixfive_all: OK (1.1s)
FatimaFaisal@xv6:~$ |
```

Memdump

```
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_
ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forp
han user/_dorphan user/_sleep user/_sixfive user/_memdump user/_find user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1102 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -
drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ memdump
Example 1:
61810
2025
Example 2:
a string
Example 3:
another
Example 4:
B60
1819438967
100
z
xyzyz
Example 5:
hello
w
o
r
l
d
$ |
```

Memdump grade test:

```
builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-b
uiltin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o kernel/console.o kernel/console.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmmodel=medany -ffreestandi
ng -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-b
uiltin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-b
uiltin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o kernel/printf.o kernel/printf.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmmodel=medany -ffreestandi
ng -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-b
uiltin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-b
uiltin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o kernel/uart.o kernel/uart.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmmodel=medany -ffreestandi
ng -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-b
uiltin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-b
uiltin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno
-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel
/main.o kernel/vm.o kernel/proc.o kernel/switch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o ker
nel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o
kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .*/ /; /'>'$d' > kernel/kernel.sym
make[1]: Leaving directory '/home/maskatia/xv6-labs-2025'
== Test memdump, examples == memdump, examples: OK (4.6s)
== Test memdump, format ii, S, p == memdump, format ii, S, p: OK (0.6s)
FatimaFaisal@xv6:~$
```

Find

```
$  
$ echo > b  
$ mkdir a  
$ echo > a/b  
$ mkdir a/aa  
$ echo > a/aa/b  
$ find . b  
./b  
./a/b  
./a/aa/b  
$
```

Find grade test:

```
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mmodel=medany  
-ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset  
-fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-  
builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprint  
f -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/printf.o kernel/printf.c  
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mmodel=medany  
-ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset  
-fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-  
builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprint  
f -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/uart.o kernel/uart.c  
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mmodel=medany  
-ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset  
-fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-  
builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprint  
f -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c  
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/st  
ring.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/  
sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysf  
ile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.  
o kernel/spinlock.o  
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions  
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm  
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym  
make[1]: Leaving directory '/home/maskatia/xv6-labs-2025'  
== Test find, in current directory == find, in current directory: OK (4.2s)  
== Test find, in sub-directory == find, in sub-directory: OK (0.7s)  
== Test find, recursive == find, recursive: OK (1.3s)  
== Test exec, recursive find == exec, recursive find: OK (1.0s)  
FatimaFaisal@xv6:~$
```

Exec

```
FatimaFaisal@xv6:~$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user
/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie
user/_logstress user/_forphan user/_dorphan user/_sleep user/_sixfive user/_memdump user/_find user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1102 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-l
egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ find . wc -exec echo hi
hi ./wc
$ sh < findtest.sh
$ $ $ $ $ hello
hello
hello
$ $ |
```

Exec grade test:

```
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany
-ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset
-fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-
builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprint
f -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/printf.o kernel/printf.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany
-ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset
-fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-
builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprint
f -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/uart.o kernel/uart.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany
-ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset
-fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-
builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprint
f -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/st
ring.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/
sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysf
ile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.
o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* // ; /^$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/maskatia/xv6-labs-2025'
== Test find, in current directory == find, in current directory: OK (4.2s)
== Test find, in sub-directory == find, in sub-directory: OK (0.7s)
== Test find, recursive == find, recursive: OK (1.3s)
== Test exec, recursive find == exec, recursive find: OK (1.0s)
FatimaFaisal@xv6:~$
```